



ME 207: Fluid Dynamics

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Lab Experiment 3: Measurement of force due to jet impingement

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Group 14

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Abstract

The aim of the experiment is to understand Reynold's transport theorem and to calculate the force exerted by a jet on the surface using the balancing weight.

1 Task 3.1

Q1: Why are liquid jets important? Briefly discuss any 3 applications of liquid jets.

Jet is a very high-pressure stream of fluid. It plays a very crucial role in industry and is used in many areas, like propulsion and cutting. Here are the three applications of jets.

Jet engines

Jets are used in jet engines to create propulsion in the vehicle. Air is compressed and mixed with fuel. After this, ignition happens, and high-speed exhaust gases are forced out of the engine, creating thrust that eventually moves the vehicle.

Water jet cutters

Jets are used in the machining industry to cut metals and other materials. An object that is supposed to be cut is subjected to water at a velocity of hundreds of meters/second. This cuts the object with a very high precision.

Liposuction surgery Jets have biomedical applications as well, such as liposuction surgery. The saline water is used to loosen the fats, which are then removed from the tissue using suction.

Q2: Read up on the Reynolds transport theorem (RTT). How is momentum conservation expressed this this theorem?

In fluid dynamics, we deal with two approaches: one is Eulerian, and the other one is Lagrangian. When we are interested in a fixed region of space, we choose control volume, which can be loosely regarded with the Eulerian approach; on the other hand, when we select a system for our analysis, which is control mass and is suitable for the Lagrangian approach. However, in many cases, we need both system and control volume for our analysis. Thus, we use Reynold's transport theorem, which relates the change of entity in the system to that in the control volume. It is stated as

$$\frac{dB_{system}}{dt} = \frac{d}{dt} \int_{CV} \rho b dV + \int_{CS} \rho b (\tilde{\mathbf{v}} \cdot \tilde{\eta}) dA, \quad (1)$$

where:

- ρ is the fluid density,
- b is the extensive property per unit mass (specific property),
- B is the total extensive property,
- $\tilde{\mathbf{v}}$ is the velocity vector,

- $\vec{\eta}$ is the outward unit normal to the control surface,
- CV refers to the control volume,
- CS refers to the control surface.

It should be noted that this equation holds true for both scalar and vector quantities. Also, if the control volume is not moving with time, the time derivative can be taken inside the equation, and the modified expression for the control volume looks like this

$$\frac{dB_{system}}{dt} = \int_{CV} \frac{\partial}{\partial t}(\rho b) dV + \int_{CS} \rho b (\vec{v} \cdot \vec{\eta}) dA, \quad (2)$$

This equation can be used to express momentum conservation as well, where momentum can be regarded as the extensive property of the system, which changes with time, while the intensive property corresponding to that is velocity.

$$\frac{d\vec{P}_{system}}{dt} = \int_{CV} \frac{\partial}{\partial t}(\rho \vec{v}) dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{\eta}) dA, \quad (3)$$

where

- $\vec{P}$ is the momentum.
- $\vec{v}$ is the velocity.

Q3: Can you compute the force exerted by a liquid jet impinging on a flat surface, using the RTT?

Reynolds transport theorem can be used to calculate the force exerted on the water. Here, the momentum can be regarded as an extensive property, and a change in momentum is equal to the force exerted on the liquid by the deflector. By Newton's third law, the same amount of force is exerted on the deflector by the jet.

Now, the force can be calculated on both deflectors using the Reynolds transport theorem. For derivation, it is assumed that the flow is steady and incompressible. **For flat deflector**

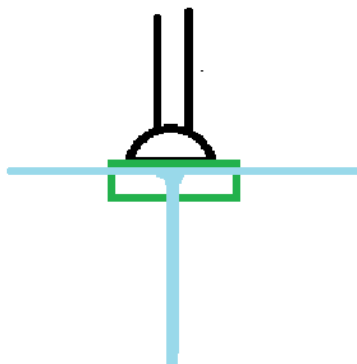


Figure 1: Control volume for the flat deflector.

Reynolds transport theorem for the momentum is given as

$$\frac{d\vec{P}_{system}}{dt} = \int_{CV} \frac{\partial}{\partial t} (\rho \vec{v}) dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{\eta}) dA, \quad (4)$$

Now, since it is a steady flow and the velocity of water crossing, the control volume does not change with time. The first term on the right-hand side is zero.

$$\frac{d\vec{P}_{system}}{dt} = \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{\eta}) dA,$$

The integral can be split into two terms

$$\frac{d\vec{P}_{system}}{dt} = \rho v_x^2 R t \int_0^{2\pi} \cos \theta d\theta \hat{i} - \rho v_y \int v_y dA \hat{j}$$

Now, the first term on the right-hand side is equal to zero because of the integral. In the second term, the velocity is constant throughout every infinitesimal component of the area. So, the equation finally becomes,

$$\frac{d\vec{P}_{system}}{dt} = -\rho A v_y^2 \hat{j}$$

Now, the derivative of the momentum of fluid in the system is the force exerted on the fluid by the deflector, and the force exerted by the jet on the deflector is the additive inverse of that derivative. Thus, the force is given by

$$\vec{F} = -\frac{d\vec{P}_{system}}{dt} = \rho A v_y^2 \hat{j}$$

where,

- v_y is the velocity of the jet.
- A is the cross-sectional area of the nozzle.

For hemispherical deflector.



Figure 2: Control volume used for hemispherical deflector

Reynolds transport theorem for the momentum is given as

$$\frac{d\vec{P}_{system}}{dt} = \int_{CV} \frac{\partial}{\partial t} (\rho \vec{v}) dV + \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{\eta}) dA, \quad (5)$$

Now, since it is a steady flow and the velocity of water crossing, the control volume does not change with time. The first term on the right-hand side is zero.

$$\frac{d\vec{P}_{system}}{dt} = \int_{CS} \rho \vec{v} (\vec{v} \cdot \vec{\eta}) dA,$$

The integral can be divided into two terms.

$$\frac{d\vec{P}_{system}}{dt} = \rho v_y \int -v_y dA \hat{j} - \rho v_y \int v_y dA \hat{j}$$

Thus, force in this case is given as:

$$\vec{F} = -\frac{d\vec{P}_{system}}{dt} = 2\rho A v_y^2 \hat{j}$$

2 Introduction



Figure 3: Image of the experimental setup

In this experiment, we are studying how water jets affect different surface shapes. The setup measures the forces created when the jet hits a target surface or stopper. One important feature is that we can change the surfaces, allowing us to test different shapes, like flat plates or parabolic curved shapes. This lets us see how the shape of the surface changes the force when the jet hits it and how different shapes affect the force created.

The system is connected to a hydraulic bench using a flexible pipe, which allows us to control the water flow and pressure. By changing things like the speed of the jet and the angle it hits, we can study how these factors affect the force on the surface or stopper. This helps us understand how the jet and the forces it creates are related. The data we gather will be useful for improving jet processes. These processes are important for things like cooling systems, surface treatments, and material processing.

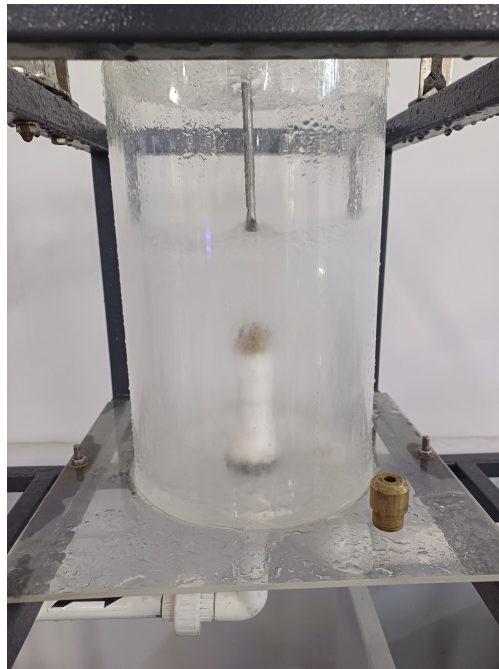


Figure 4: Deflection of the jet beam by deflector

3 Objectives

- (i) Study the force generated by a liquid jet impinging on a surface by adjusting a balancing weight.
- (ii) Observe the variations in the force for different jet speeds.
- (iii) Interpret the results using the Reynolds transport theorem.

4 Experimental Procedure

The following steps were taken to perform the experiment:

1. Before beginning the experiment, we will need to note down the weight of different masses placed on the horizontal rod, including the weight of the horizontal rod; their weight has been written on the side.
2. The weight of the flat fulcrum and the rod connected to it is unknown, and first, we have to find their mass.
3. The horizontal rod can be balanced by adjusting the weights on both sides; now, because the system is in equilibrium, we can use the moment balance equation about the hinged point to calculate the mass of the connecting rod and fulcrum.
4. The next step is to turn on the motor and adjust the water's flow rate, which will be transferred through the connecting pipes and come out of the nozzle.
5. Because the area of the nozzle cross-section is much less than the pipe, the velocity of water coming out increases, creating a high-speed jet.
6. We have to ensure that the water does not come out of the apparatus from the small hole on the top, and one of the members has to manually hold the system horizontally for the same.
7. The next step is to find the system's equilibrium position, which is achieved by manually changing the position of masses on both sides of the rod.
8. After the equilibrium position has been identified, we have to take the center of mass (COM) reading of these masses from the hinged position, which is the zero position on the scale.
9. Using the moment balance equation about the hinged point, we can calculate the force due to jet impingement as the net moment is zero for the hinged point.
10. The above steps must be repeated for the same flow rate with the curved fulcrum.
11. The next step is to calculate the flow rate of the jet, for which we will calculate the time the water takes to fill a particular height in the sink tank in which a narrow transparent tube is attached with markings.
12. We must ensure that the sink's bottom valve is closed and that the flow rate entering inside the cylinder in which the jet throws out water equals the flow rate exiting the cylinder.
13. The time water takes to fill a particular height is measured, and multiple readings are to be taken to minimize the error.
14. As we know the area of the sink tank, the time it takes to fill a particular height, and the nozzle area, we can use Bernoulli's theorem to calculate the velocity of water exiting the jet.
15. The same experiment will be repeated for a different flow rate, controlled by adjusting the valve.

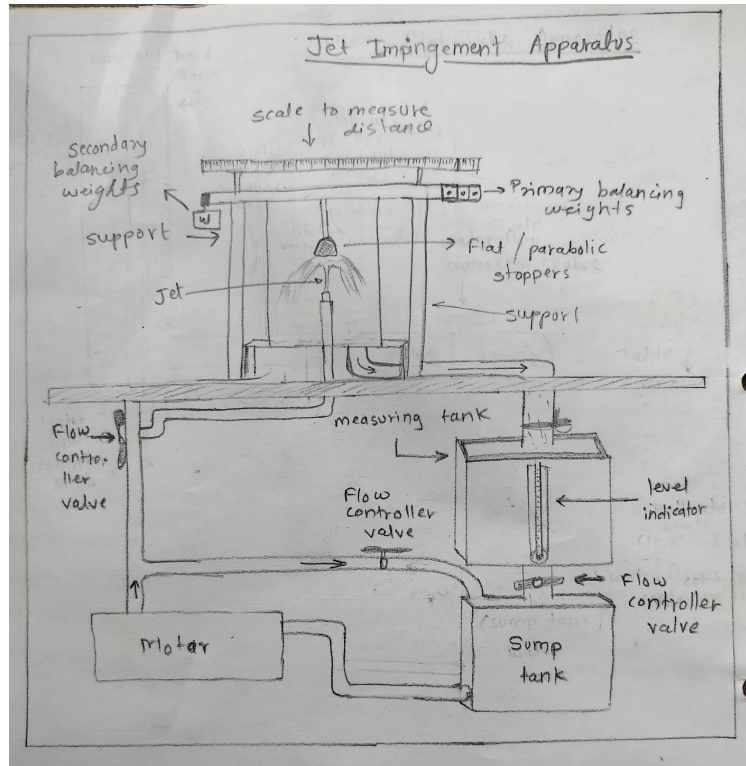


Figure 5: Schematic diagram of the apparatus

5 Results and Error Analysis

5.1 Experimental Results

For this experiment, we have following values:

Parameter	Value
Mass of the cylinder	1.84 kg
Sum of small masses	0.3307 kg
Position of center of mass (cylinder)	-20.5 cm
Position of center of mass (small masses)	36 cm
Mass of the long horizontal rod	0.689 kg
Position of center of mass (long rod)	14.5 cm

Table 1: Measured Experimental Parameters

$$\sum (F_x d) = 0$$

$$(1.84g \cdot 20.5 - 0.3307g \cdot 36 - 0.689g \cdot 11 - m_{\text{fulcrum+vertical rod}} \cdot g \cdot 14.5) = 0$$

$$m_{\text{fulcrum+vertical rod}} = \frac{1.84 \cdot 20.5 - 0.3307 \cdot 36 - 0.689 \cdot 11}{14.5}$$

$$m_{\text{fulcrum+vertical rod}} = 1.258 \text{ kg}$$

This value is incorrect as the actual mass of the vertical rod and fulcrum is less than 500 g. The corrected values, measured using a weighing machine, are:

Component	Measured Mass (kg)
Vertical rod	0.053
Flat fulcrum	0.0439
Curved fulcrum	0.053
Large rod	0.678
Large weight (cylinder)	1.0825

Table 2: Corrected Mass Values

For Flat Fulcrum:-

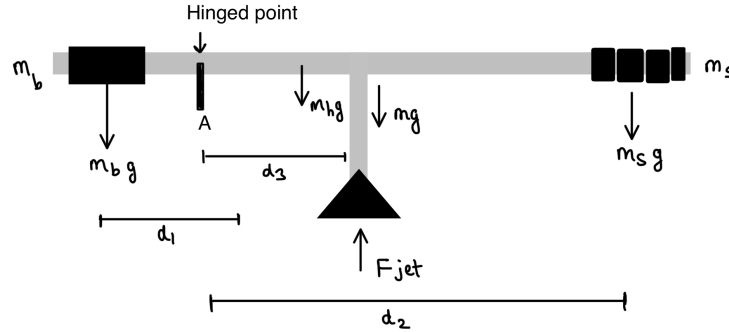


Figure 6: Free body diagram for the flat deflector.

Applying the Moment balance equation,

$$\Sigma F_x d = 0$$

$$F_1 d_1 - F_2 d_2 - F_3 d_3 - F_4 d_4 + F_{jet} d_1 = 0$$

$$m_b g d_1 - m_s g d_2 - m g d_3 - m_h g d_4 + F_{jet} d_1 = 0$$

$$F_{jet} d_3 = -m_b g d_4 + m_s g d_2 + m g d_3 + m_h g d_1$$

Here,

- Big mass (cylinder): $m_b = 1.0825 \text{ kg}$, (As given by the TA)
- Sum of small masses: $m_s = 0.3307 \text{ kg}$,

- Mass of the rod (vertical) + mass of fulcrum(flat): $m = 0.0969 \text{ kg}$
- Mass of horizontal big rod: $m_h = 0.678 \text{ kg}$
- g (Gravity due to acceleration) $= 9.81 \text{ m/s}^2$
- d_1 = distance of cylinder from the hinged point,
- d_2 = distance of the small mass from the hinged point,
- d_3 = distance of m from hinged point $= 0.145 \text{ m}$,
- d_4 = distance of Big rod's COM(center of mass) from hinged point $= 0.115 \text{ m}$,
- F_1 = force by the cylinder,
- F_2 = force by the small mass,
- F_3 = force by m ,
- F_4 = Force by the big rod,
- F_{jet} = force by water jet

We have two values of flow rate, we'll find the jet flow for both flow rate.
For the first flow rate:

d_1	d_2	F_{jet}
0.205 m	0.399 m	0.139 N
0.205 m	0.398 m	0.117 N
0.206 m	0.402 m	0.133 N

For the second flow rate

d_1	d_2	F_{jet}
0.140m	0.245m	0.1454N
0.141m	0.244m	1.358N
0.140m	0.243m	1.409N

F_{jet} for the first flow rate	$F_{jet,avg}$ for the second flow rate
$(0.130 \pm 0.008N)$	$1.407 \pm 0.032N$

For the Curved Fulcrum -

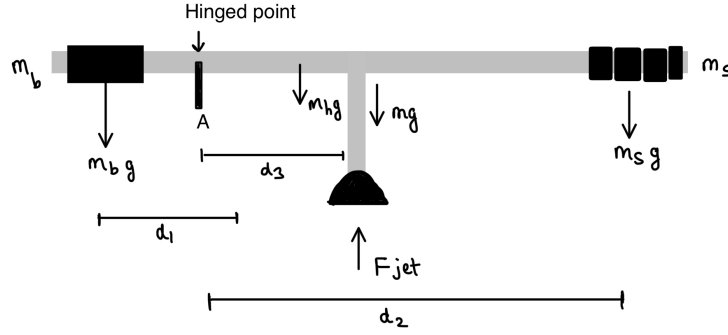


Figure 7: Free body diagram for the hemispherical deflector.

Applying the Moment balance equation,

$$\Sigma F_x d = 0$$

$$F_1 d_1 - F_2 d_2 - F_3 d_3 - F_4 d_4 + F_{jet} d_1 = 0$$

$$F_{jet} d_3 = -m_b g d_4 + m_s g d_2 + m g d_3 + m_h g d_1$$

Where,

- Mass of rod (horizontal big rod): $m_h = 678$ gm
- Mass of the rod (vertical) + mass of fulcrum (Curved) : $m = 110$ gm
- Big mass (cylinder): $m_b = 1082.5$ gm,
- Small mass: $m_s = 330.7$ gm,
- g (Gravity due to acceleration) $= 9.81 \text{ m/s}^2$
- d_1 = distance of cylinder from the hinged point,
- d_2 = distance of the small mass from the hinged point,
- d_3 = distance of m from the hinged point
- d_4 = distance of Big rod's COM (center of mass) from hinged point $= 11.5$ cm
- F_1 = force by the cylinder,
- F_2 = force by the small mass,
- F_3 = force by m ,
- F_4 = Force by the big rod
- F_{jet} = force by water jet

For the first flow rate:-

d_1	d_2	F_{jet}
0.205 m	0.431 m	0.983 N
0.205 m	0.430 m	0.961 N
0.206 m	0.433 m	0.955 N

For the second flow rate :-

d_1	d_2	F_{jet}
0.141 m	0.32 m	3.187 N
0.138 m	0.319 m	3.384 N
0.139 m	0.32 m	3.334 N

F_{jet} for the first flow rate	$F_{jet,avg}$ for the second flow rate
$(0.966 \pm 0.011)N$	$(3.302 \pm 0.076)N$

Task 3.2:-

(a) Measure the balancing weight and its position for different flow rates. From this calculate the force exerted on the vane by the liquid jet.

This is done in the experimental results.

(b) Compute the theoretical force using the RTT. Compare the measured force and the theoretically calculated force.

The derivation is written in the 3.1 Question:-3.

(i) Flat fulcrum $F_{theoretical} = \rho A_n (v_n)^2$ Where,

- A_n = area of nozzle,
- v_n = velocity of fluid coming from nozzle

But we do not know the velocity of fluid from the nozzle. So, we'll use the continuity equation. $A_n v_n = A_t v_t$ where,

- A_t = area of tank = 0.16 m²
- V_t = velocity of fluid from a tank

For the first flow rate of the nozzle,

From above equations, the theoretical value of force in the case of flat fulcrum will be,

$$F_{theoretical} = \frac{\rho (A_t v_t)^2}{A_n}$$

Reading number	Time taken	height gained	v_t	$F_{theoretical}$
1	18.94 s	2 cm	0.00106 m/s	1.014 N
2	20.64 s	2 cm	0.00097 m/s	0.849 N
3	19.46 s	2 cm	0.00103 m/s	0.958 N
4	19.89 s	2 cm	0.00101 m/s	0.921 N
5	20.22 s	2 cm	0.00099 m/s	0.885 N
Average	19.58 s	2 cm	0.00102 m/s	0.939 N

$F_{experimental}$	$F_{theoretical}$ for the second flow rate
$(0.130 \pm 0.008)N$	$(0.939 \pm 0.057)N$

For the second flow rate of the nozzle,
 From above equations, the theoretical value of force in the case of flat fulcrum will be,

$$F_{theoretical} = \frac{\rho(A_t v_t)^2}{A_n}$$

Reading number	Time taken	height gained	v_t	$F_{theoretical}$
1	20.16 s	3 cm	0.001488 m/s	1.999 N
2	20.45 s	3 cm	0.001467 m/s	1.943 N
3	18.9 s	3 cm	0.001587 m/s	2.274 N
4	19.83 s	3 cm	0.001513 m/s	2.067 N
5	18.93 s	3 cm	0.001585 m/s	2.268 N
Average	19.545 s	3 cm	0.001528 m/s	2.108 N

$F_{experimental}$	$F_{theoretical}$ for the second flow rate
$(1.407 \pm 0.032)N$	$(2.108 \pm 0.137)N$

(ii) Curved fulcrum

To find the $F_{theoretical}$ for the curved fulcrum, the method is continued which is used in the flat fulcrum.

So, for the first flow rate

$$F_{theoretical} = \frac{2\rho(A_t v_t)^2}{A_n}$$

Reading number	Time taken	height gained	v_t	$F_{theoretical}$
1	18.94 s	2 cm	0.00106 m/s	2.029 N
2	20.64 s	2 cm	0.00097 m/s	1.699 N
3	19.46 s	2 cm	0.00103 m/s	1.916 N
4	19.89 s	2 cm	0.00101 m/s	1.843 N
5	20.22 s	2 cm	0.00099 m/s	1.770 N
Average	19.58 s	2 cm	0.00102 m/s	1.851 N

$F_{experimental}$	$F_{theoretical}$ for the second flow rate
$(0.966 \pm 0.011)N$	$(1.851 \pm 0.103)N$

For the second flow rate.

$$F_{theoretical} = \frac{2\rho(A_t v_t)^2}{A_n}$$

Reading number	Time taken	height gained	v_t	$F_{theoretical}$
1	20.16 s	3 cm	0.001488 m/s	3.999 N
2	20.45 s	3 cm	0.001467 m/s	3.887 N
3	18.9 s	3 cm	0.001587 m/s	4.549 N
4	19.83 s	3 cm	0.001513 m/s	4.135 N
5	18.93 s	3 cm	0.001585 m/s	4.538 N
Average	19.545 s	3 cm	0.001528 m/s	4.222 N

$F_{\text{experimental}}$	$F_{\text{theoretical}}$ for the second flow rate
$(3.302 \pm 0.076)N$	$(4.222 \pm 0.256)N$

5.2 Error Analysis

(c) **Report experimental uncertainties and quantify errors by running the experiments multiple times.** The percentage error was calculated using:

$$\text{Percentage error} = \left| \frac{F_{\text{experimental}} - F_{\text{theoretical}}}{F_{\text{theoretical}}} \right| \times 100\%$$

Fulcrum type	Flow rate number	$F_{\text{experimental}}$	$F_{\text{theoretical}}$	Percentage error
Flat fulcrum	Flow rate 1	0.130 N	0.939 N	86.15 %
	Flow rate 2	1.407 N	2.108 N	33.25 %
Curved Fulcrum	Flow rate 1	0.966 N	1.851 N	47.81 %
	Flow rate 2	3.302 N	4.222 N	21.79 %

5.3 Experimental uncertainties

During the experiment, If the mass balance occurs, I move the large mass about 2-4 cm or the small masses together about 3-5 cm. For that, balance is not disturbed, which shows the error in the apparatus because it has to disturb and change with the position. To measure the volume of the sump tank for the flow rate, we have less time to measure the height of the sump tank because when we close the valve, the water increases in the sump tank and disturbs the flow of water, which balances the mass. So, this causes the error.

(d) **In case there are mismatches between observed and the calculated forces, can you specify some reasons which may cause these mismatches?**

The mismatch between calculated(theoretical) and observed(experimental) values can happen due to some reasons. In theoretical calculations, sometimes approximate values are taken into account. in this case, forces are round off to three integers, so it is possible that it can create a mismatch between theoretical and experimental values. Also, some environmental factors like air resistance can create a mismatch because we're not considering air friction in theoretical calculations. The location of COM(center of mass) is not always exact at the center every time, so the moment can't be exactly zero by our approximation sometimes. So, material defects can be the reason for the mismatch between the two results.

6 Observation

Comparing the values of the force of the jet using theoretical and experimental methods. The force of the jet calculated theoretically is not the same as the one found experimentally. This will be the same theoretically and experimentally if the accurate apparatus is used. The next observation we made is that the force of jet is more due to the curved fulcrum than the flat fulcrum, theoretically the force should be double but experimentally due to errors that factor varied a lot.

7 Conclusion

In this experiment, after finding the values of jet forces using two methods(one is theoretical and one is experimental), RTT(Reynold's transfer theorem) can be an approximation to find jet flows by different flow rates and fulcrums. However, for more accurate results, other factors such as measurement error, instrument error, and dynamic effects should be taken into consideration.

8 Acknowledgements

We would like to thank our Teaching Assistant Mr.Akash for explaining the concept of Reynolds transport theorem.

References

- [1] Yunus A. and John M., *Fluid Mechanics: Fundamentals and Applications*, McGraw Hill, 4th edition, 2018.
- [2] Wikipedia contributors, "Water jet cutter," Wikipedia, Feb. 03, 2025. Available: https://en.wikipedia.org/wiki/Water_jet_cutter
- [3] Classroom slides from Prof. Udiptta Ghosh and Prof. Dilip Sundaram